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Semiconductor device and method for fabricating the same

Abstract

A method for fabricating semiconductor device includes the steps of forming a first inter-metal dielectric (IMD) layer, forming two via holes and a trench in the first IMD layer, forming a metal layer in the two via holes and the trench for forming a metal interconnection and a spin orbit torque (SOT) layer, forming a magnetic tunneling junction (MTJ) on the SOT layer, forming a first hard mask on the MTJ, forming a second hard mask on the first hard mask, forming a cap layer adjacent to the MTJ, and forming a second IMD layer around the cap layer.

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202118101	12/2020	TW	N/A

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The invention relates to a method for fabricating semiconductor device, and more particularly to a method for fabricating magnetoresistive random access memory (MRAM).

2. Description of the Prior Art

(2) Magnetoresistance (MR) effect has been known as a kind of effect caused by altering the resistance of a material through variation of outside magnetic field. The physical definition of such effect is defined as a variation in resistance obtained by dividing a difference in resistance under no magnetic interference by the original resistance. Currently, MR effect has been successfully utilized in production of hard disks thereby having important commercial values. Moreover, the characterization of utilizing GMR materials to generate different resistance under different magnetized states could also be used to fabricate MRAM devices, which typically has the advantage of keeping stored data even when the device is not connected to an electrical source.

(3) The aforementioned MR effect has also been used in magnetic field sensor areas including but not limited to for example electronic compass components used in global positioning system (GPS) of cellular phones for providing information regarding moving location to users. Currently, various magnetic field sensor technologies such as anisotropic magnetoresistance (AMR) sensors, GMR sensors, magnetic tunneling junction (MTJ) sensors have been widely developed in the market. Nevertheless, most of these products still pose numerous shortcomings such as high chip area, high cost, high power consumption, limited sensibility, and easily affected by temperature variation and

how to come up with an improved device to resolve these issues has become an important task in this field.

SUMMARY OF THE INVENTION

(4) According to an embodiment of the present invention, a method for fabricating semiconductor device includes the steps of forming a first inter-metal dielectric (IMD) layer, forming two via holes and a trench in the first IMD layer, forming a metal layer in the two via holes and the trench for forming a metal interconnection and a spin orbit torque (SOT) layer, forming a magnetic tunneling junction (MTJ) on the SOT layer, forming a first hard mask on the MTJ, forming a second hard mask on the first hard mask, forming a cap layer adjacent to the MTJ, and forming a second IMD layer around the cap layer.

(5) According to another aspect of the present invention, a semiconductor device includes a first inter-metal dielectric (IMD) layer on a substrate and a metal interconnection and a spin orbit torque (SOT) layer in the first IMD layer. Preferably, top surfaces of the first IMD layer and the SOT layer are coplanar.

(6) These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. **1-8** illustrate a method for fabricating a MRAM device according to an embodiment of the present invention,

(2) FIGS. **9-11** illustrate a method for fabricating a MRAM device according to an embodiment of the present invention.

DETAILED DESCRIPTION

(3) Referring to FIGS. **1-8**, FIGS. **1-8** illustrate a method for fabricating a MRAM device according to an embodiment of the present invention. As shown in FIG. **1**, a substrate **12** made of semiconductor material is first provided, in which the semiconductor material could be selected from the group consisting of silicon (Si), germanium (Ge), Si—Ge compounds, silicon carbide (SiC), and gallium arsenide (GaAs), and a MRAM region **14** and a logic region **16** are defined on the substrate **12**.

(4) Active devices such as metal-oxide semiconductor (MOS) transistors, passive devices, conductive layers, and interlayer dielectric (ILD) layer **18** could also be formed on top of the substrate **12**. More specifically, planar MOS transistors or non-planar (such as FinFETs) MOS transistors could be formed on the substrate **12**, in which the MOS transistors could include transistor elements such as gate structures (for example metal gates) and source/drain region, spacer, epitaxial layer, and contact etch stop layer (CESL). The ILD layer **18** could be formed on the substrate **12** to cover the MOS transistors, and a plurality of contact plugs could be formed in the ILD layer **18** to electrically connect to the gate structure and/or source/drain region of MOS transistors. Since the fabrication of planar or non-planar transistors and ILD layer is well known to those skilled in the art, the details of which are not explained herein for the sake of brevity.

(5) Next, a metal interconnect structure **20** is formed on the ILD layer **18** to electrically connect the aforementioned contact plugs, in which the metal interconnect structure **20** includes an inter-metal dielectric (IMD) layer **24** and metal interconnections **26** embedded in the IMD layer **24**. In this embodiment, each of the metal interconnections **26** from the metal interconnect structure **20** preferably includes a trench conductor, in which each of the metal interconnections **26** could be embedded within the IMD layer **24** according to a single damascene process or dual damascene process. For instance, each of the metal interconnections **26** could further include a barrier layer **34**

and a metal layer **36**, in which the barrier layer **34** could be selected from the group consisting of titanium (Ti), titanium nitride (TiN), tantalum (Ta), and tantalum nitride (TaN) and the metal layer **36** could be selected from the group consisting of tungsten (W), copper (Cu), aluminum (Al), titanium aluminide (TiAl), and cobalt tungsten phosphide (CoWP). Since single damascene process and dual damascene process are well known to those skilled in the art, the details of which are not explained herein for the sake of brevity.

(6) Next, a stop layer **28** and an IMD layer **30** is formed on the metal interconnect structure **20**, and one or more etching process is conducted by using a patterned mask (not shown) to remove part of the IMD layer **30** and part of the stop layer **28** for forming an opening **72** exposing the metal interconnection **26** underneath. Preferably, the opening **72** includes a trench **74** or trench opening and two via holes **76** connecting to the bottom of the trench **74**.

(7) Next, as shown in FIG. 2, a barrier layer **34** and a metal layer **36** is formed to fill the opening **72** completely, and a planarizing process such as chemical mechanical polishing (CMP) process is conducted to remove part of the metal layer **36**, part of the barrier layer **34**, and part of the IMD layer **30** so that the top surfaces of the remaining barrier layer **34**, metal layer **36**, and IMD layer **30** are coplanar. It should be noted that the planarized barrier layer **34** and metal layer **36** together constitute a metal interconnection **32** and a spin orbit torque (SOT) layer **38** serving as a channel for SOT MRAM in the opening **72**, in which the barrier layer **34** and metal layer **36** filled in the lower two via holes **76** become the metal interconnection **32** made of via conductors while the barrier layer **34** and metal layer **36** filled in the trench **74** opening become the SOT layer **38**.

(8) Similar to the aforementioned embodiment, the barrier layer **34** from the metal interconnection **32** and SOT layer **38** could be selected from the group consisting of titanium (Ti), titanium nitride (TiN), tantalum (Ta), and tantalum nitride (TaN) and the metal layer **36** could be selected from the group consisting of tungsten (W), copper (Cu), aluminum (Al), titanium aluminide (TiAl), and cobalt tungsten phosphide (CoWP). In this embodiment, the metal layer **36** in the metal interconnection **32** and SOT layer **38** is preferably made of tungsten (W) and the stop layer **28** is preferably made of nitrogen doped carbide (NDC), silicon nitride, silicon carbon nitride (SiCN), or combination thereof.

(9) Next, as shown in FIG. 3, a MTJ stack **40** or stack structure, a hard mask **68**, and another hard mask **42** are formed on the SOT layer **38**. In this embodiment, the formation of the MTJ stack **40** could be accomplished by sequentially depositing a pinned layer, a barrier layer, and a free layer on the SOT layer **38**. Preferably, the pinned layer could be made of ferromagnetic material including but not limited to for example iron, cobalt, nickel, or alloys thereof such as cobalt-iron-boron (CoFeB) or cobalt-iron (CoFe). Alternatively, the pinned layer could also be made of antiferromagnetic (AFM) material including but not limited to for example ferromanganese (FeMn), platinum manganese (PtMn), iridium manganese (IrMn), nickel oxide (NiO), or combination thereof, in which the pinned layer is formed to fix or limit the direction of magnetic moment of adjacent layers. The barrier layer could be made of insulating material including but not limited to for example oxides such as aluminum oxide (AlO_{sub}.x) or magnesium oxide (MgO). The free layer could be made of ferromagnetic material including but not limited to for example iron, cobalt, nickel, or alloys thereof such as cobalt-iron-boron (CoFeB), in which the magnetized direction of the free layer could be altered freely depending on the influence of outside magnetic field.

(10) Preferably, the hard mask **68** includes conductive material or metal such as ruthenium (Ru) and the hard mask **42** preferably includes conductive or dielectric material such as tantalum (Ta), tantalum nitride (TaN), titanium (Ti), titanium nitride (TiN), platinum (Pt), copper (Cu), gold (Au), aluminum (Al), or combination thereof.

(11) Next, as shown in FIG. 4, an etching process or more specifically a photo-etching process is conducted to pattern the hard mask **42** for exposing the surface of the hard mask **68** underneath. Specifically, the photo-etching process could be accomplished by first forming a patterned mask

(not shown) such as patterned resist on the hard mask **42**, and then an etching process is conducted by using the patterned mask to remove part of the hard mask **42** for forming a patterned hard mask **42** and exposing the surface of the hard mask **68**. Preferably, the etching process conducted at this stage includes a reactive ion etching (RIE) process.

(12) Next, as shown in FIG. 5, one or more etching process such as an ion beam etching (IBE) process is conducted to remove part of the hard mask **68** and part of the MTJ stack **40** to form a MTJ **48** on the MRAM region **14**. Preferably, part of the hard mask **42** may be consumed during the etching process so that the overall thickness of the hard mask **42** could be slightly reduced. Next, a cap layer **50** is formed on the MTJ **48** to cover the surface of the IMD layer **30** on the MRAM region **14** and logic region **16**. In this embodiment, the cap layer **50** preferably includes silicon nitride (SiN), but could also include other dielectric material including but not limited to for example silicon oxide, silicon oxynitride (SiON), or silicon carbon nitride (SiCN).

(13) Next, an etching process could be conducted without forming any patterned mask to remove part of the cap layer **50** so that the cap layer **50** directly contacting the top surface of the hard mask **42** and the cap layer **50** directly contacting the top surface of the SOT layer **38** have same thickness while the cap layer **50** directly contacting the top surface of the SOT layer **38** and the cap layer **50** directly contacting the sidewall of the MTJ **48** have different thicknesses, or more specifically the thickness the cap layer **50** directly contacting the top surface of the SOT layer **38** is less than the thickness of the cap layer **50** directly contacting the sidewall of the MTJ **48**. In this embodiment, the thickness of the cap layer **50** adjacent to or directly contacting sidewall of the MTJ **48** is approximately twice or more such as three or even four times the thickness of the cap layer **50** on top or directly contacting the top surface of the SOT layer **38**.

(14) Next, as shown in FIG. 6, a photo-etching process is conducted to pattern the cap layer **50** by using a patterned mask (not shown) such as a patterned resist as mask to remove part of the cap layer **50** through etching process and expose the top surface of the IMD layer **30**, in which the sidewall of the patterned cap layer **50** could be aligned with or not aligned with the sidewall of the SOT layer **38** underneath. If the sidewall of the patterned cap layer **50** were not aligned with the sidewall of the SOT layer **38**, the sidewall of the cap layer **50** could be overlapping or not overlapping the SOT layer **38**. For instance, if the cap layer **50** were formed to extend outward as the width of the cap layer **50** is greater than the width of the SOT layer **38**, the bottom surface of the cap layer **50** would contact the top surface of the SOT layer **38** and the IMD layer **30** at the same time. Alternatively, if the cap layer **50** were formed to slightly shrink inward as the width of the cap layer **50** is less than the width of the SOT layer **38**, the bottom surface of the cap layer **50** would only contact the top surface of the SOT layer **38** directly but not contacting the top surface of the IMD layer **30**, which are all within the scope of the present invention.

(15) Next, as shown in FIG. 7, an atomic layer deposition (ALD) process is conducted to form an ID layer **52** on the cap layer **50** and the IMD layer **30**, in which the ID layer **52** could include an ultra low-k (ULK) dielectric layer including but not limited to for example porous material or silicon oxycarbide (SiOC) or carbon doped silicon oxide (SiOCH).

(16) Next, a planarizing process such as chemical mechanical polishing (CMP) process or etching back process is conducted to remove part of the IMD layer **52** so that the top surface of the remaining IMD layer **52** includes a planar surface and is still higher than the top surface of the cap layer **50**. Next, a pattern transfer process is conducted by using a patterned mask (not shown) to remove part of the ID layer **52**, part of the IMD layer **30**, and part of the stop layer **28** on the MRAM region **14** and logic region **16** to form contact holes (not shown) exposing the metal interconnections **26** underneath and conductive materials are deposited into the contact hole afterwards. For instance, a barrier layer selected from the group consisting of titanium (Ti), titanium nitride (TiN), tantalum (Ta), and tantalum nitride (TaN) and metal layer selected from the group consisting of tungsten (W), copper (Cu), aluminum (Al), titanium aluminide (TiAl), and cobalt tungsten phosphide (CoWP) could be deposited into the contact holes, and a planarizing

process such as CMP could be conducted to remove part of the conductive materials including the aforementioned barrier layer and metal layer to form metal interconnections **58** in the contact holes electrically connecting the metal interconnections **26**. It should be noted that each of the metal interconnections **58** could include a trench conductor disposed in the IMD layer **52** and via conductor disposed in the IMD layer **30**, in which the bottom surface of the trench conductor or top surface of the via conductor is even with the top surface of the SOT layer **38** or bottom surface of the MTJ **48**.

(17) Next, as shown in FIG. **8**, a stop layer **60** is formed on the MRAM region **14** and logic region **16** to cover the IMD layer **52** and metal interconnections **58**, an IMD layer **62** is formed on the stop layer **60**, and one or more photo-etching process is conducted to remove part of the IMD layer **62**, part of the stop layer **60**, part of the IMD layer **52**, and part of the cap layer **50** on the MRAM region **14** and logic region **16** to form contact holes (not shown). Next, conductive materials are deposited into each of the contact holes and a planarizing process such as CMP is conducted to form metal interconnections **64** connecting the MTJ **48** and metal interconnections **58** underneath, in which the metal interconnections **64** on the MRAM region **14** directly contact the hard mask **42** and metal interconnections **58** underneath while the metal interconnections **64** on the logic region **16** directly contacts the metal interconnections **58** on the lower level.

(18) In this embodiment, the stop layers **60** and **28** could be made of same or different materials, in which the two layers **60**, **28** could all include nitrogen doped carbide (NDC), silicon nitride, silicon carbon nitride (SiCN), or combination thereof. Similar to the metal interconnections formed previously, each of the metal interconnections **64** could be formed in the IMD layer **62** through a single damascene or dual damascene process. For instance, each of the metal interconnections **64** could further include a barrier layer and a metal layer, in which the barrier layer could be selected from the group consisting of titanium (Ti), titanium nitride (TiN), tantalum (Ta), and tantalum nitride (TaN) and the metal layer could be selected from the group consisting of tungsten (W), copper (Cu), aluminum (Al), titanium aluminide (TiAl), and cobalt tungsten phosphide (CoWP). Since single damascene process and dual damascene process are well known to those skilled in the art, the details of which are not explained herein for the sake of brevity. This completes the fabrication of a semiconductor device according to an embodiment of the present invention.

(19) Referring to FIGS. **9-11**, FIGS. **9-11** illustrate a method for fabricating a MRAM device according to an embodiment of the present invention. As shown in FIG. **9**, it would be desirable to first follow the process conducted in FIG. **1** by forming an IMD layer **18** on a substrate **12** and then forming metal interconnect structures **20**, **22** to electrically connect the contact plugs in the IMD layer **18**, in which the metal interconnect structure **20** includes an IMD layer **24** and metal interconnections **26** embedded in the IMD layer **24** while the metal interconnect structure **22** includes a stop layer **28**, an IMD layer **30**, and at least two metal interconnections **32** embedded in the stop layer **28** and the IMD layer **30**.

(20) In this embodiment, each of the metal interconnections **26** from the metal interconnect structure **20** preferably includes a trench conductor and the metal interconnection **32** from the metal interconnect structure **22** on the MRAM region **14** includes a via conductor. Preferably, each of the metal interconnections **26**, **32** from the metal interconnect structures **20**, **22** could be embedded within the IMD layers **24**, **30** and/or stop layer **28** according to a single damascene process or dual damascene process. For instance, each of the metal interconnections **26**, **32** could further include a barrier layer **34** and a metal layer **36**, in which the barrier layer **34** could be selected from the group consisting of titanium (Ti), titanium nitride (TiN), tantalum (Ta), and tantalum nitride (TaN) and the metal layer **36** could be selected from the group consisting of tungsten (W), copper (Cu), aluminum (Al), titanium aluminide (TiAl), and cobalt tungsten phosphide (CoWP). Since single damascene process and dual damascene process are well known to those skilled in the art, the details of which are not explained herein for the sake of brevity. In this embodiment, the metal layers **36** in the metal interconnections **26** are preferably made of copper, the metal layer **36** in the metal

interconnections **32** is made of tungsten, the IMD layers **24, 30** are preferably made of silicon oxide such as tetraethyl orthosilicate (TEOS), and the stop layer **28** is preferably made of nitrogen doped carbide (NDC), silicon nitride, silicon carbon nitride (SiCN), or combination thereof. Next, another IMD layer **78** is formed on the metal interconnect structure **22**, and a photo-etching process is conducted to remove part of the IMD layer **78** for forming an opening **80** or trench exposing the metal interconnection **32** underneath.

(21) Next, as shown in FIG. **10**, a barrier layer **34** and a metal layer **36** are formed to fill the opening **80** completely, and then a planarizing process such as CMP is conducted to remove part of the barrier layer **34**, part of the metal layer **36**, and part of the IMD layer **78** so that the top surfaces of the remaining barrier layer **34** and metal layer **36** are even with the top surface of the IMD layer **78** to form a SOT layer **38** in the opening **80**.

(22) Similar to the aforementioned embodiment, the SOT layer **38** preferably serves a channel for the SOT MRAM device as the barrier layer could be selected from the group consisting of titanium (Ti), titanium nitride (TiN), tantalum (Ta), and tantalum nitride (TaN) and the metal layer **36** could include tantalum (Ta), tungsten (W), platinum (Pt), or hafnium (Hf) and/or topological insulator such as bismuth selenide (Bi.sub.xSe.sub.1-x). It should be noted that in contrast to the metal layer **36** in the metal interconnection **32** and the metal layer **36** in the SOT layer **38** from the aforementioned embodiment are made of same material such as tungsten, the metal layer **36** in the metal interconnection **32** and the metal layer **36** in the SOT layer **38** in this embodiment could be made of same or different material depending on the demand of the product. In this embodiment, the metal layers **36** from the metal interconnection **32** and SOT layer **38** are both made of tungsten and the IMD layers **24, 30, 78** are made of silicon oxide such as TEOS, but not limited thereto.

(23) Next, as shown in FIG. **11**, processes conducted in FIGS. **3-8** could then be carried out to form a MTJ stack **40**, a hard mask **68**, and another hard mask **42** on the SOT layer **38**, pattern the hard mask **42**, the hard mask **68**, and the MTJ stack **40** to form a MTJ **48**, form a cap layer **50** on the MTJ **48**, pattern the cap layer **50** so that the edge of the cap layer **50** could be aligned or not aligned with sidewall of the SOT layer **38** underneath, form an IMD layer **52** on the cap layer **50**, form metal interconnections **58** in the IMD layers **52, 78, 30** to connect to the metal interconnections **26**, form a stop layer **60** and IMD layer **62** on the IMD layer **52**, and then form metal interconnections **64** in the IMD layer **62** to electrically connect the MTJ **48** and metal interconnections **58**, in which the metal interconnection **64** on the MRAM region **14** directly contacts the hard mask **42** underneath while the metal interconnection **64** on the logic region **16** contacts the lower level metal interconnection **58**.

(24) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A method for fabricating a semiconductor device, comprising: forming a first inter-metal dielectric (IMD) layer on a substrate; forming two via holes and a trench in the first IMD layer; and forming a single barrier layer and a single metal layer in the two via holes and the trench for forming a metal interconnection and a spin orbit torque (SOT) layer at the same time in the two via holes and the trench, wherein a topmost surface of the SOT layer is even with a top surface of the first IMD layer.
2. The method of claim 1, further comprising: forming a magnetic tunneling junction (MTJ) on the SOT layer; forming a first hard mask on the MTJ; forming a second hard mask on the first hard mask; forming a cap layer adjacent to the MTJ; and forming a second IMD layer around the cap layer.

3. The method of claim 2, further comprising forming the cap layer on the SOT layer.
 4. The method of claim 2, further comprising forming the second IMD layer on the SOT layer.
 5. The method of claim 2, wherein top surfaces of the first IMD layer and the SOT layer are coplanar.
 6. The method of claim 2, wherein the first hard mask comprises ruthenium (Ru).
 7. The method of claim 2, wherein the second hard mask comprises metal nitride.
 8. The method of claim 1, wherein the metal interconnection and the SOT layer comprise same material.
 9. The method of claim 1, wherein the metal interconnection and the SOT layer comprises tungsten.
 10. A semiconductor device, comprising: a first inter-metal dielectric (IMD) layer on a substrate; a second IMD layer on the first IMD layer; a metal interconnection in the second IMD layer; a third IMD layer on the second IMD layer; a spin orbit torque (SOT) layer in the third IMD layer, wherein each of the metal interconnection and the SOT layer is made of a single barrier layer and a single metal layer and a topmost surface of the SOT layer is even with a top surface of the third IMD layer; a magnetic tunneling junction (MTJ) on the SOT layer; a first hard mask on the MTJ; a second hard mask on the first hard mask; a cap layer adjacent to the MTJ; and a fourth IMD layer around the cap layer.
 11. The semiconductor device of claim 10, wherein bottom surfaces of the third IMD layer and the SOT layer are coplanar.
 12. The semiconductor device of claim 10, wherein the metal interconnection and the SOT layer comprise same material.
 13. The semiconductor device of claim 10, wherein the metal interconnection and the SOT layer comprises tungsten.
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